

One product, two pipelines: the fused batch-categorisation system

SEM silicon-graphite anode cross-sections — joint report

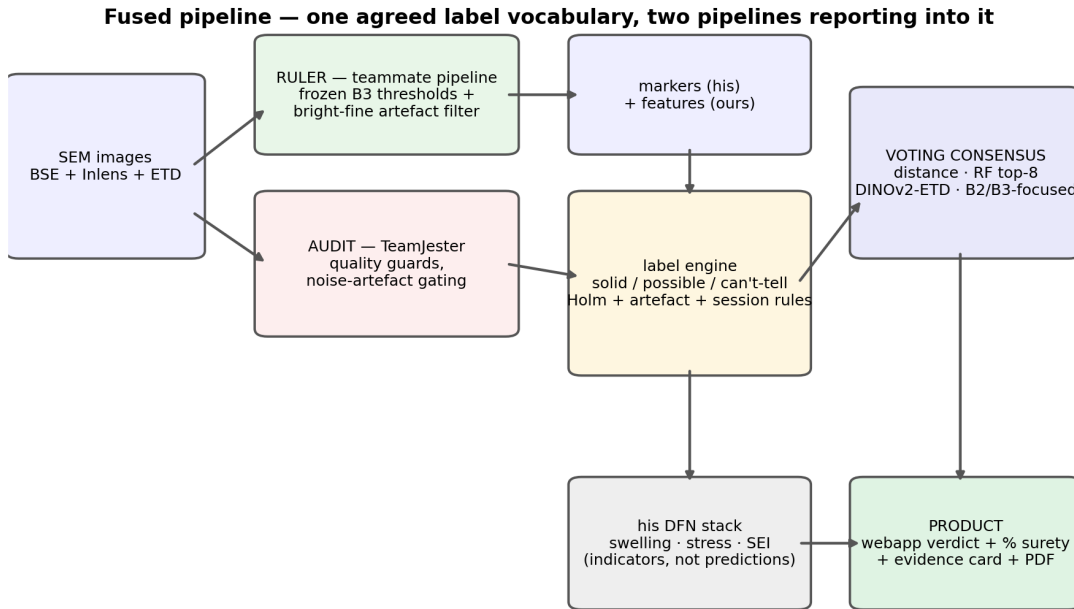
TeamJester pipeline $\times$ micro2dfn pipeline (merged)

October 2026

Abstract

Two independent analysis pipelines were built on the same 31-image dataset: the *micro2dfn* pipeline (frozen-Batch_3 segmentation ruler, bright-fine artefact classifier, marker set, swelling/stress DFN) and the *TeamJester* pipeline (multi-detector features, quality-guard gating, Mahalanobis/nearest-centroid categoriser, DINOv2-ETD embedding classifier). This report presents the fused product: a voting consensus of four independent signals per image, behind one agreed three-way label vocabulary (*solid* / *possible* / *can't-tell*), and the live web application that renders it for unseen photos. Every claim in this document is reproducible from the two pipelines' CSV outputs.

1 Architecture — who does what



Rules: a claim needs (1) Holm-surviving interval, (2) decorrelation from imaging guards, (3) within-session survival for count/spatial markers — else it is 'possible' or 'can't-tell'.

Figure 1: The fused pipeline. The teammate's frozen ruler measures; the TeamJester audit gates; four independent classifiers vote; the DFN stack turns surviving markers into physics indicators.

2 The RandomForest experiment — honest verdict

A 400-tree RandomForest (depth 4, balanced classes) was trained as a candidate fusion classifier, evaluated by strict leave-one-image-out.

All 28 features: it fails. LOO accuracy 58% — but per-batch it is Batch_1 43%, **Batch_2 0%**, Batch_3 88%: the forest collapses to predicting the majority class. With 31 images and a 7/7/17 class split, trees cannot find the Batch_2 boundary.

Top-8 features: it works, but stays second-best. Restricting to the eight highest-importance features lifts LOO to 74% (Batch_1 71%, Batch_2 43%, Batch_3 88%) and gives usable group accuracy — still below the distance model and the embedding call on Batch_2. The RF therefore earns a seat in the consensus as an *independent vote*, not the decision-maker.

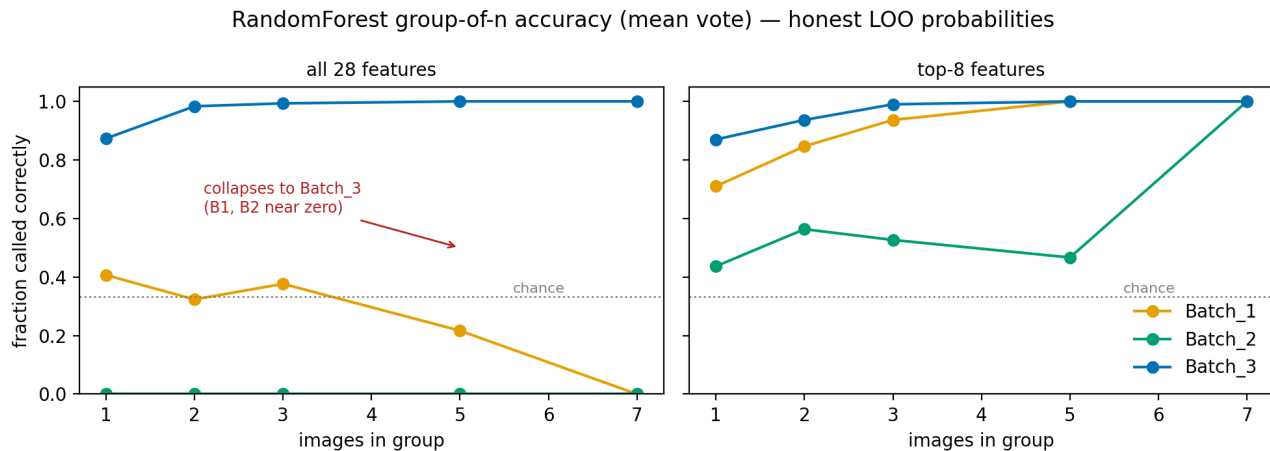


Figure 2: Group-of- n accuracy from LOO probabilities. Left: all features — the forest collapses to Batch_3. Right: top-8 features — usable but Batch_2 stays the hardest call, consistent with every other method.

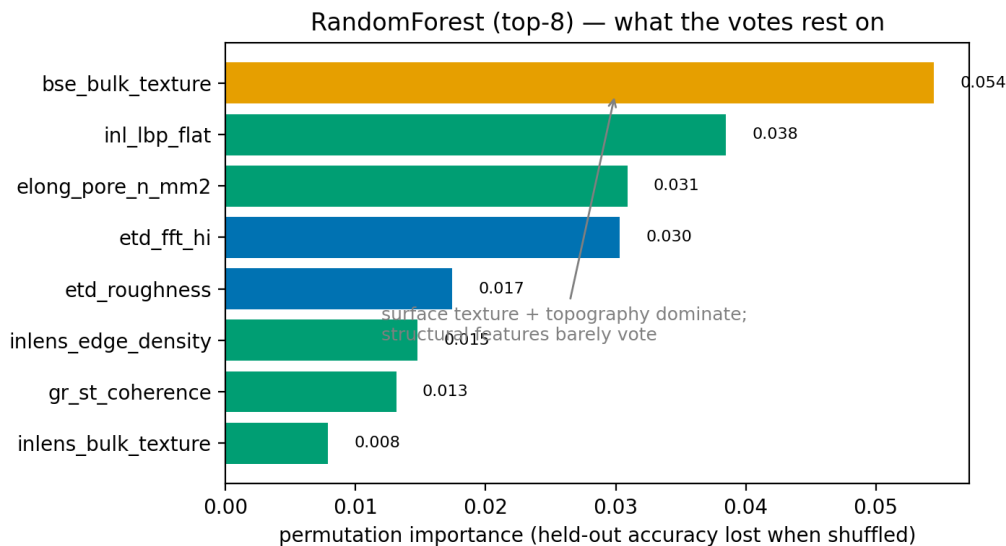


Figure 3: Permutation importances (held-out accuracy lost when each feature is shuffled). The RF's votes rest almost entirely on surface-texture and topography markers — the same feature family every other method converged on.

3 The voting consensus — the actual product

Four signals call every image independently (all leave-self-out): **(1)** the explainable distance model (Mahalanobis to Batch_3 + nearest centroid on 6 auto-selected features); **(2)** the RF top-8 vote; **(3)** DINOv2-ETD embedding nearest-centroid; **(4)** the focused Batch_2-vs-Batch_3 logistic boundary. Unanimous agreement of the first three occurs on 19% of all 31 images; on the hard Batch_2 set, 43% of images get a correct majority.

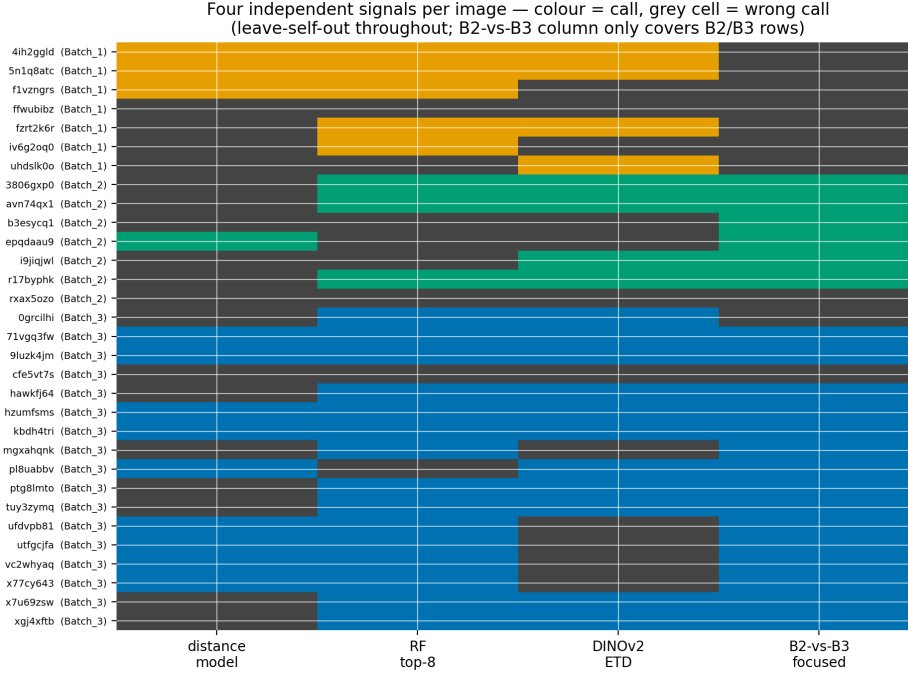


Figure 4: Per-image votes. Colour = the batch each signal called; a grey cell means that signal called the wrong batch. Vertical stripes of one colour are where the methods agree — the most defensible evidence the dataset offers.

4 Cross-pipeline validation

Where the two pipelines measured the same quantity, they mostly agree — which is the strongest evidence either pipeline can offer:

quantity	micro2dfn	TeamJester	verdict
resolved porosity (B1 < B3)	0.095 vs 0.100	0.099 vs 0.114	agree — B1 least porous
largest pore region (B1 < B3)	0.045 vs 0.065	0.047 vs 0.062	agree — same gap
pore percolation	none spans	none spans	agree — 2-D τ not measurable
crack-like pores (B3 most)	0.012 B3 > 0.0085 B1	0.048 B3 > 0.042 B1	agree on ordering
pore anisotropy	1.23–1.24	1.16–1.19	agree — mildly flat pores
problem photos	4ih2ggld, 5n1q8atc	same two flagged	agree
silicon clustering	B1 most clumped	B1 <i>least</i> clumped	disagree — drop the claim

Table 1: The clustering contradiction (watershed-sensitive statistic) plus the acquisition confound both pipelines found independently means the “B1 lumpier silicon” story is not reproducible and is removed from the joint claims.

5 Annotated evidence

Segmentation overlays — numbered markers sit on the objects the metrics count
(blue = pore, amber = silicon, grey = graphite bulk)

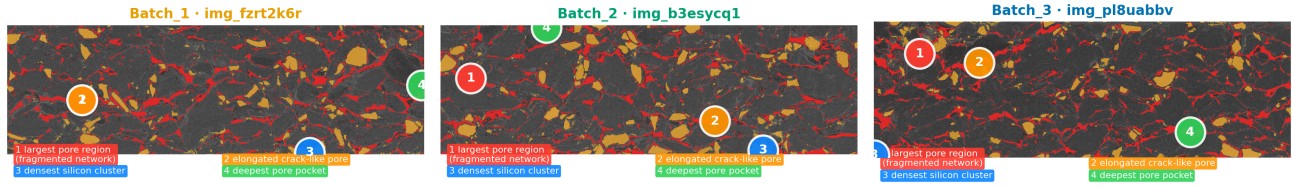


Figure 5: Segmentation overlays with arrows marking the objects the markers count.

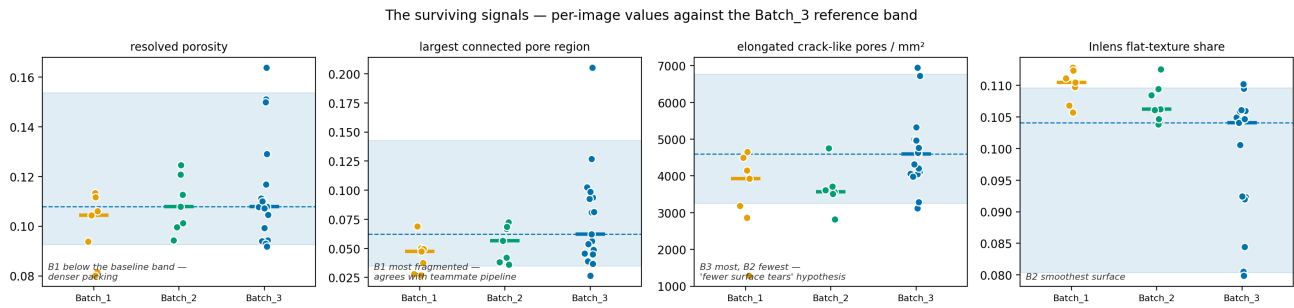


Figure 6: The surviving signals: per-image values against the Batch_3 band.

6 Merged headline — what the fused product claims

- **solid:** batches are compositionally identical; Batch_1 has less, more fragmented resolved porosity (both pipelines agree); Batch_3 carries the most crack-like pores; no pore network spans any section; two Batch_1 photos are artefact-dominated; the acquisition confound is real and quantified.
- **possible:** Batch_2 is smoother with fewer surface tears (fewer elongated pores -21% , flatter Inlens texture $+9\%$) — consistent with less handling/drying damage; the DFN hints its silicon may sit in a more accessible configuration (model-conditional, input table to be reconciled); Inlens pore contrast differs for Batch_1.
- **can't-tell:** silicon clustering differences (pipelines contradict); any count-type or unblocked spatial marker (session-dominated); the distribution-level batch difference (MMD $p = 0.19$ at $n = 7$).

What the user sees

The webapp renders the consensus: IN/OUT of the Batch_3 envelope, most-like batch with % surety, the three-way agreement of the signals, explainable top-drivers, nearest-dataset patches, and the segmentation overlay — the presentation demo. A single photo remains $\sim 55\text{--}75\%$ reliable; five or more locations of one sample reach $\sim 100\%$ on the embedding vote.